VS-05 Statistical Product Composition

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VS-05.0 Statistical Product Composition

Purpose and Scope

This document specifies the complete business layer for the **Statistically composed and composite statistical products** business domain, corresponding to the *VS-05.0 Statistical Product Composition Value Stream*. It also covers 4 governance process types:

- **Design** — how composite products, indicators, and coherence checks are specified and planned (produces metadata)
- **Implementation** — how the composition pipeline is built, configured, and tested (produces the execution process)
- **Execution** — how outputs are operationally integrated, composed, checked, and registered (produces statistical data)
- **Management** — how the domain is monitored, quality-assessed, and improved (produces quality reports)

> **Out of Scope**
>
> Application and Technology layer specifications are out of scope for this document. Re-estimation, weighting, and model-based analysis are out of scope — they belong to VS-04.0 Estimation, Analysis, and Statistical Modeling; VS-05 composes existing validated outputs, it does not re-estimate them. Publishing, distribution, and user services are out of scope — they belong to VS-06.0 Dissemination and Service Delivery.

A note on numbering: each `VS-0x.y` below is both a value stream stage (*what value* is added) and a business process (*how* it is done); the numbering maps one-to-one. See the key-concepts box above.

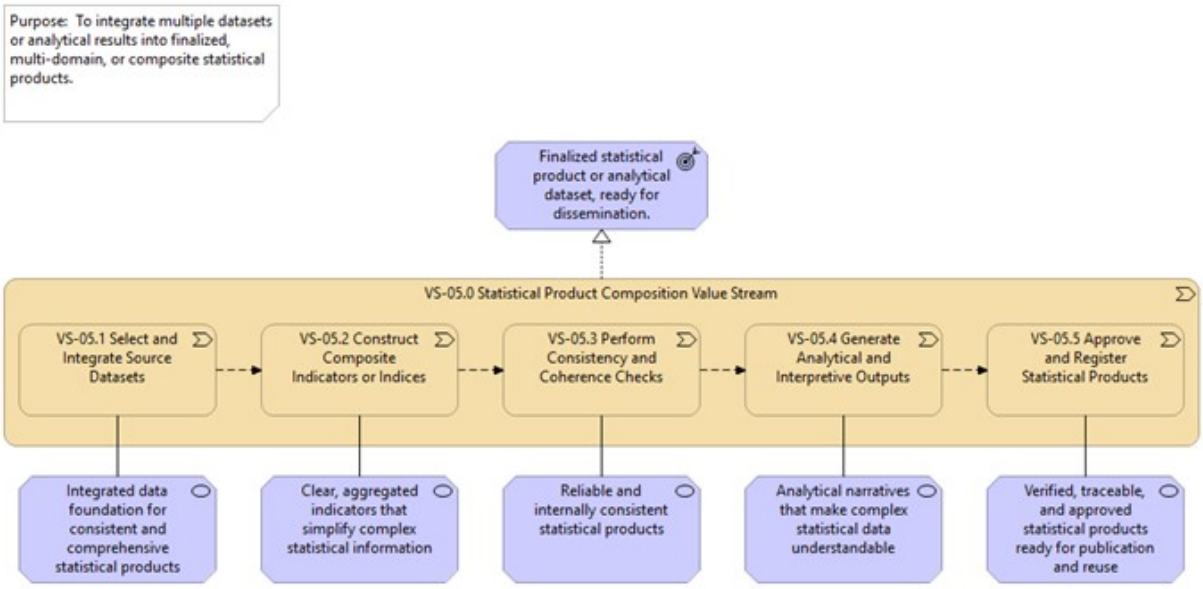


Figure 1: VS-05.0 Statistical Product Composition: purpose, stages and outcomes

Position in the Statistical Value Chain

VS-05.0 is the fifth primary value stream — it receives Output Data (validated statistics) from VS-04.0 and brings multiple analytical results together into coherent, high-value statistical products ready for dissemination. It is the "Compose & Approve Products" orchestration stage where separate analyses become finished statistical products.

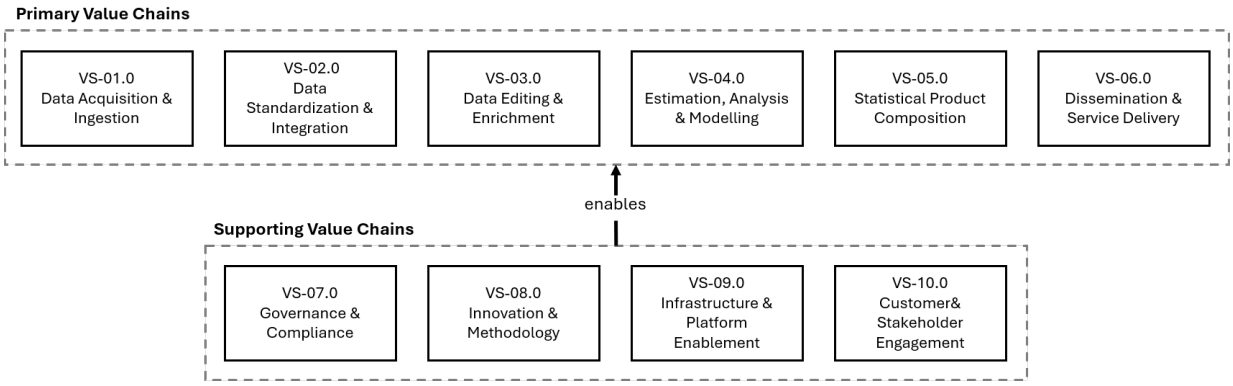


Figure 2: Value Stream Context Map. VS-05.0 is the composition stage that turns validated analytical outputs into approved statistical products.

Steady State Transition

VS-05.0 operates **within Steady State #4 (Output Data / Statistics)**. It receives individual validated analytical outputs (the estimation form of State #4 produced by VS-04.0) and produces **approved, registered composite statistical products** (the final-product form of State #4) — coherent across sources and time, validated against dissemination and confidentiality standards, and ready to hand to dissemination. State #4 is shared between the estimation domain (VS-04) and this composition domain (VS-05).

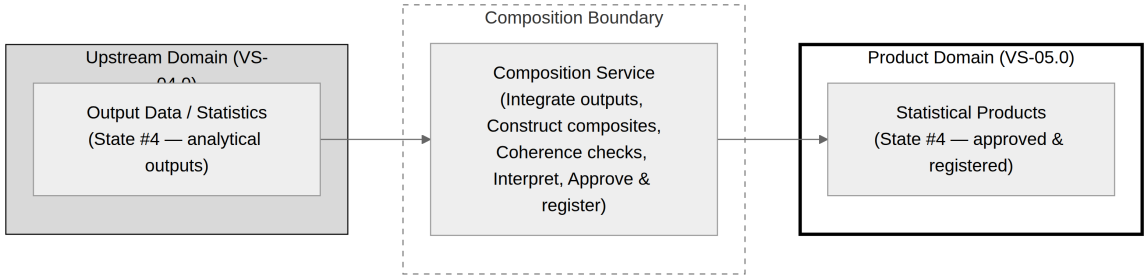


Figure 3: Steady State Transition within State #4: from individual analytical outputs to approved, registered composite statistical products.

Product approval criteria (Examples)*

- Source outputs selected and integrated with their accuracy/methodology metadata preserved
- Composite indicators, indices, and time series constructed per documented specifications
- Internal consistency and coherence with related and prior products verified
- Analytical and interpretive content (commentary, visualizations) produced and reviewed
- Product validated against dissemination and confidentiality standards
- Approved by the designated authority and registered with a unique product identifier
- Full lineage from the product back to its component State #4 outputs and the composition method

The composition discipline — coherence and interpretability

VS-05 does not re-estimate; it **composes** validated outputs into products. The guarantee here is **coherence and interpretability**: composite products are internally consistent, methodologically sound, coherent with related and prior statistics, and accompanied by the metadata and interpretation users need. Every product carries lineage to its component outputs and the composition method, and is **approved against dissemination and confidentiality standards** before it is registered.

Inherited from VS-04.0

VS-05 receives Output Data **plus** the metadata VS-04 attached — methodology, weights, accuracy measures (sampling error, confidence intervals), and the record of the disclosure control already applied. VS-05 **propagates** this accuracy and quality metadata into the composite product (it does not discard or recompute it), uses it to weight or caveat composed figures, and runs coherence checks across the combined outputs. Composition never silently changes a component estimate; any adjustment for coherence is documented and traceable.

The Four-Process Governance Pattern

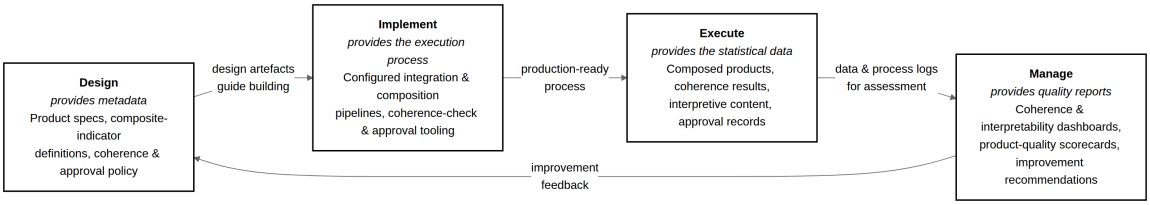


Figure 4: The Four-Process Governance Pattern. Each process type produces a distinct artefact class. The cycle closes when Management findings feed back into Design improvements.

Governing Principles

The following principles directly govern the VS-05.0 business domain:

Principle	Statement	Application to VS-05.0
SAF-P01	Actively participate in open-source communities to maintain collective control over statistical processes	NSIs should contribute to open source communities for composition, indicator-construction, and reporting/visualization tooling, ensuring collective control over critical product capabilities rather than vendor dependence.
SAF-P07	Implement loosely coupled processes and systems	The composition domain must be independent from upstream estimation and downstream dissemination. The approved statistical product serves as a decoupling interface — dissemination accesses it without knowledge of how it was composed.
SAF-P08	Data is shared property	Approved statistical products and their components, once registered, are available to all authorized consumers. Composition should reuse existing validated outputs rather than duplicate analysis.

Principle	Statement	Application to VS-05.0
SAF-P09	No data without metadata and classification	Every composite product carries metadata: its component outputs and their accuracy, the composition/indicator methodology, coherence-check results, classification, and the approval record. This is what makes products interpretable and trustworthy.
SAF-P10	Security By Design	Access controls, integrity verification, and a tamper-evident composition and approval record are embedded in the pipeline so that every product is attributable and reproducible, not added as an afterthought.
SAF-P11	Privacy By Design	Composition is validated against confidentiality standards, and the disclosure risk of composite products (including the risk of combining outputs) is assessed by design, before registration.

i Cross-cutting principle application

Several principles cut across all four process types. SAF-P09 (no data without metadata and classification) underpins interpretability: Design defines the product and required metadata, Execute captures component lineage, composition method, and coherence results, and Manage assesses product quality. SAF-P10 (Security By Design) governs the tamper-evident approval/composition record across Execution (Sections 5.1–5.5). SAF-P11 (Privacy By Design) applies especially to approval (Section 5.5), where the composite product's confidentiality and disclosure risk are confirmed before registration. SAF-P07 (loosely coupled) governs both the State #4 intake interface from VS-04.0 and the handover interface to VS-06.0.

Capability Foundations

VS-05.0 draws on the following SAF capabilities:

Capability Group	Capabilities	Role in VS-05.0
1.1 Statistical design and build	1.1.2 Design statistics product, 1.1.3 Build statistics product	Design the statistical products and composite indicators, and build the composition pipeline
2.1 Data management	2.1.1 Data lifecycle management, 2.1.2 Metadata management, 2.1.3 Statistical quality management, 2.1.4 Statistical register management	Manage products, metadata, and quality; assess coherence; integrate across registers and time
3.2 Computerization and automation	3.2.1 Application development, 3.2.2 Functional management	Build and maintain integration, indicator-construction, and reporting/visualization engines
3.3 Security and Privacy	3.3.1 Privacy management, 3.3.2 Security planning	Validate products against confidentiality standards and assess composite disclosure risk

Capability Group	Capabilities	Role in VS-05.0
4.1 Governance	4.1.1 Strategy and governance, 4.1.2 Policy and planning, 4.1.4 Improve management, 4.1.5 Accountability	Govern the domain, steward product approval, plan improvements, report on compliance

Metadata Perspective

It is considered that metadata is an essential product of the Design phase and a critical concern of every other phase. In this domain, metadata is what makes a composite product interpretable: each product must carry its components, methodology, coherence results, and approval. The table below consolidates the metadata produced across all four process types, covering both data metadata (about the products) and process metadata or paradata (about how the process runs).

Process Type	Data Metadata Produced	Process Metadata (Paradata) Produced
Design	Product specifications, composite-indicator definitions, coherence criteria, approval & confidentiality policy	Design decisions log, methodology version, review and approval records
Implement	Integration configurations, indicator-construction parameters, coherence-check parameters, product templates	Build log, test results, coherence-test outcomes, deployment approvals
Execute	Per-product component lineage, composition/indicator method, coherence-check results, interpretive content, approval record	Operator identity, timing, software/version, approval workflow trail
Manage	Coherence & interpretability assessments, product-quality scores, lineage assessments, trend analyses	Coherence dashboards, product-quality trends, improvement decisions, audit findings

Business Roles and Actors

Role	Process Orientation	Description
Upstream Data Provider (VS-04.0)	External (internal NSI)	Supplies Output Data (State #4) with methodology/accuracy metadata to the composition domain
Standards Body	External	Maintains product, indicator, and exchange standards the NSI conforms to (e.g., SDMX, index manuals)
Product Manager	Design	Designs the statistical product portfolio, product specifications, and release content
Indicator/Index Methodologist	Design	Specifies composite indicators, indices, and time-series construction methods
Coherence Designer	Design	Designs the consistency and coherence checks across sources and time
Data Engineer	Implementation	Builds and configures integration, indicator-construction, and reporting engines and the pipeline
System Configurator	Implementation	Configures composition services, output-source connections, and access controls

Role	Process Orientation	Description
Test Manager	Implementation	Plans and executes testing of the composition pipeline (technical, coherence, reproducibility)
Process Owner	Management	Accountable for the end-to-end performance of the composition domain
Compliance Officer	Management	Ensures confidentiality and standards compliance of products, produces audit reports
Quality Manager	Management	Monitors coherence/interpretability KPIs, produces quality reports, and drives continuous improvement
Integration Specialist	Execution	Selects and integrates the component outputs into the product structure
Subject-Matter Author	Execution	Produces analytical and interpretive content, commentary, and visualizations
Approval Authority	Execution	Validates products against standards and approves them for registration and release
Data Steward	Execution	Registers approved products and versions, manages metadata entries, and applies access controls

Design Processes — "provides metadata"

Framework alignment: GSBPM Phases 1–2 — see Framework Mappings.

The design phase produces the **metadata artefacts** that govern all subsequent phases — the product specifications, composite-indicator definitions, coherence checks, and the approval and confidentiality policy. Without well-designed product specs, composition becomes ad-hoc and incoherent, and management has no baseline to measure coherence and interpretability against.

Design the Statistical Product Portfolio

The starting point is defining what products will be produced. This sub-process specifies each statistical product — its purpose, target users, content, structure, breakdowns, release frequency, and the set of component State #4 outputs it draws on. The team designs how component outputs are selected and integrated (which estimates, which domains, which time periods) and how their accuracy metadata is carried into the product, aligned with GSBPM 2.5 (Design processing and analysis).

Activity	Key Outputs	Framework Ref
Specify each product: purpose, users, content, structure, breakdowns, frequency	Product Specification	GSBPM 2.1
Define the component outputs each product draws on and selection rules	Component-Selection Specification	GSBPM 2.5
Define how component accuracy/quality metadata is carried into the product	Metadata-Propagation Rules	SAF-P09
Define product classification and versioning identity	Product Metadata Standard	SAF-P09

NSI Example — Labour Market Publication:

The labour-market quarterly publication is specified to combine the LFS employment-rate estimates (national and NUTS-2), the unemployment series, and administrative job-vacancy figures into a single coherent release, carrying each component's confidence intervals into the published tables.

Design Composite Indicators and Indices

This sub-process specifies the derived measures the products present — composite indicators, indices, ratios, and time-series aggregates. Each is defined by its formula, component inputs, weighting/aggregation method, base period, and chaining/linking rules for time series, together with the metadata and classification it carries. Where a composite combines outputs of differing accuracy, the design specifies how uncertainty is represented, aligned with GSBPM 2.5.

Activity	Key Outputs	Framework Ref
Specify composite indicators/indices: formula, inputs, weighting, base period	Composite-Indicator Specification	GSBPM 2.5
Specify time-series construction: chaining, linking, benchmarking	Time-Series Specification	GSBPM 2.5
Define how component uncertainty is represented in composites	Uncertainty-Representation Rules	SAF-P09
Define metadata and classification for each composite	Composite Metadata Standard	SAF-P09

Design Consistency and Coherence Checks

This sub-process designs the checks that ensure integrated products are sound. It specifies internal-consistency checks (totals reconcile to components, breakdowns sum to totals), coherence checks against related statistics (e.g., national accounts, prior products) and across time, and the tolerance thresholds and disposition rules (accept, flag, return for review). The revision policy for products is also designed here, aligned with GSBPM 2.5 and GAMSO Quality Management.

Activity	Key Outputs	Framework Ref
Define internal-consistency checks (totals, breakdowns reconcile)	Consistency Check Set	GSBPM 2.5
Define coherence checks vs. related statistics and across time	Coherence Check Set	GAMSO QM
Define tolerance thresholds and disposition rules	Coherence Tolerance Policy	GSBPM 2.5
Define product revision policy	Revision Policy	GAMSO QM

Design Analytical and Interpretive Content

This sub-process designs the human-facing content that makes products interpretable. It specifies the analytical narrative and commentary structure, the visualizations and tables, the key messages and their evidence, and the editorial/quality-review workflow for interpretive content. Accessibility and plain-language standards are designed in, aligned with GSBPM 2.5.

Activity	Key Outputs	Framework Ref
Specify analytical narrative and commentary structure	Content Template	GSBPM 2.5
Specify visualizations and tables	Visualization Specification	GSBPM 2.5
Define key messages and evidence requirements	Key-Message Guidelines	GSBPM 2.5
Define editorial and quality-review workflow for content	Editorial Review Policy	GAMSO QM

Design Approval, Confidentiality, and Versioning Framework

The final design sub-process ensures products are properly approved, protected, and traceable. It defines the approval workflow and authority, the confidentiality/disclosure standards a product must meet before registration (including the risk of combining outputs), the product-versioning policy (identity, revision triggers, immutability, retention), the lineage requirements from each product to its component outputs, and the role-based access controls in line with SAF-P10.

Activity	Key Outputs	Framework Ref
Define approval workflow and authority	Approval Policy	SAF 4.1.5
Define confidentiality/disclosure standards for products (incl. combination risk)	Confidentiality Standard	SAF-P11
Define product-versioning and lineage requirements	Versioning & Lineage Policy	CSDA Traceability
Design role-based access control for products	Access Control Design	SAF-P10

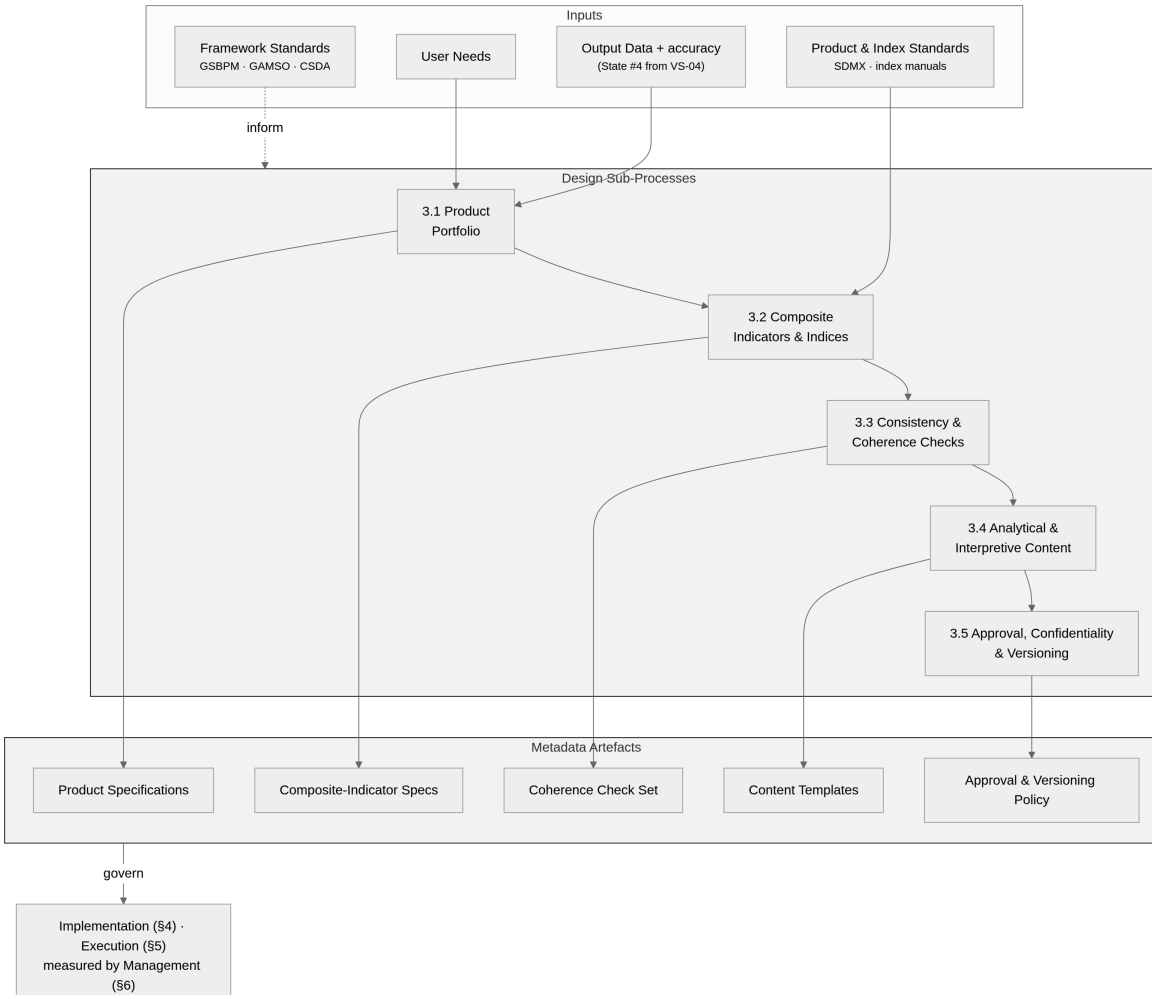


Figure 5: Design Process Decomposition. Output data with accuracy, user needs, and product standards flow into five design sub-processes, producing metadata artefacts that govern Implementation, Execution, and Management.

Implementation Processes — "provides the execution process"

Framework alignment: GSBPM Phase 3 (Build) — see Framework Mappings.

The implementation phase takes the metadata artefacts from Design and **builds, configures, and tests** the production-ready composition pipeline. Nothing runs operationally until this phase is complete.

Build Integration and Composition Engines

This sub-process turns the product and integration designs into operational tools. The integration engine is built or configured to select and combine the designed component outputs into the product structure, carrying their accuracy/quality metadata. The composition engine is built to assemble products from their components per the specifications. Output-source connections are configured to the State #4 catalog with version pinning so the exact component versions used are recorded. All composition logic is version-controlled in line with SAF-P10, following GSBPM 3.1–3.2.

Activity	Key Outputs	Framework Ref
Build/configure the integration engine (select and combine component outputs)	Configured integration engine	GSBPM 3.2
Build/configure the composition engine (assemble products from components)	Configured composition engine	GSBPM 3.2
Configure pinned connections to the State #4 output catalog	Output-source integration	CSDA Metadata Mgmt
Version-control all composition logic	Versioned composition repository	SAF-P10

NSI Example — Business Statistics Product:

The SBS composite product engine is configured to integrate the register-based aggregates and the survey-based estimates for the same NACE × region domains, pinning the exact `OD-BUS` output versions and carrying their accuracy metadata into the combined tables.

Build Indicator-Construction and Time-Series Engines

The designed composite-indicator and time-series specifications are implemented. Indicator/index calculators are built from the designed formulae and weighting, time-series engines are configured for chaining, linking, and benchmarking, and uncertainty representation is implemented so composites carry an appropriate measure of accuracy. Every composite output is produced with its method and inputs attached, following GSBPM 3.2.

Activity	Key Outputs	Framework Ref
Build indicator/index calculators from designed formulae	Indicator-construction engine	GSBPM 3.2
Configure time-series chaining, linking, and benchmarking	Time-series engine	GSBPM 3.2
Implement uncertainty representation for composites	Composite-accuracy component	SAF-P09
Attach method and inputs to every composite	Composite metadata binding	SAF-P09

Build Coherence-Check, Content, and Approval Tooling

This sub-process implements the consistency/coherence checks, the content-production tooling, and the approval/versioning store. Coherence tooling implements the designed internal-consistency and cross-source/time checks with their tolerances. Content tooling supports the analytical narrative, tables, and visualizations per the templates, with the editorial review workflow. The approval-and-versioning store is built so each approved product version is immutable, identified, reproducible from its components plus method, and accompanied by its lineage and approval record. Event notifications are configured to alert downstream consumers (VS-06.0) when a product is approved and registered.

Activity	Key Outputs	Framework Ref
Implement coherence/consistency checks with tolerances	Coherence tooling	GSBPM 3.2
Build content/visualization tooling with editorial workflow	Content tooling	GSBPM 3.2
Build the approval-and-versioning store (immutable, identified, reproducible)	Approval & versioning store	CSDA Traceability
Configure "Statistical Product Registered" event for downstream consumers	Event publication	GSBPM 3.2

Test and Validate the Pipeline

Before the pipeline goes into production, it must be rigorously verified. Technical testing covers all components — integration, composition, indicator/time-series engines, coherence checks, content tooling, and the approval store (GSBPM 3.5). Integration testing validates the end-to-end flow from output intake through composition, coherence checks, content, approval, to registration. A **coherence test** confirms composites reconcile to components and cohere with prior products; a **reproducibility test** confirms a product regenerates identically from its component versions plus the composition method. The phase concludes with finalization of the production system and documentation of operational procedures (GSBPM 3.6–3.7).

Activity	Key Outputs	Framework Ref
Technical testing of all individual components	Unit test reports	GSBPM 3.5
Integration testing: end-to-end flow from output intake to registration	Integration test report	GSBPM 3.5
Coherence test (composites reconcile to components and prior products)	Coherence test report	GSBPM 3.6
Reproducibility test (regenerate a product from components + method)	Reproducibility report	GSBPM 3.6
Finalize production system and document operational procedures	Operational Procedures Manual, Production-Ready Pipeline	GSBPM 3.7

Execution Processes — "provides the statistical data"

Framework alignment: the later GSBPM phases for this domain — see Framework Mappings.

The execution phase runs the **designed and implemented pipeline** operationally for each production cycle. This corresponds to the five VS-05 value stream stages. In a mature steady state, much of the work has been pre-designed (Section 3) and pre-built (Section 4) — execution focuses on running the cycle,

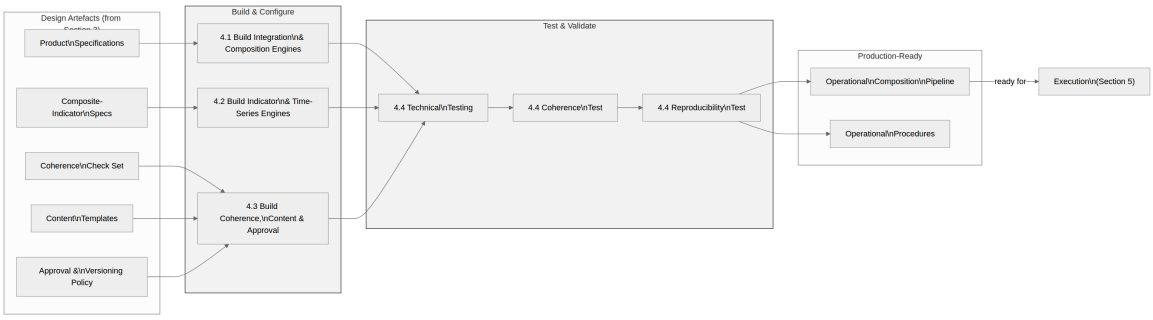


Figure 6: Implementation Pipeline from design artefacts through build/configure/test to production-ready system.

composing products, checking coherence, and capturing component lineage and approval so that every product is coherent, interpretable, and reproducible.

Select and Integrate Source Datasets

Value created: *The right validated outputs brought together as the basis for a coherent product.*

When an "Output Data Registered" event is received from VS-04.0, execution begins by selecting and integrating the component outputs for the product. Per-cycle execution applies the designed selection rules to pick the correct State #4 output versions (by domain, period, and accuracy), integrates them into the product structure, and carries their accuracy/methodology metadata forward. Components from multiple domains, surveys, or time periods are combined per the specification, aligned with GSBPM 7.1/7.2.

Activity	Key Outputs	Framework Ref
Select the correct component output versions per the specification	Selected components	GSBPM 7.1
Integrate components into the product structure	Integrated product base	GSBPM 7.2
Carry component accuracy/methodology metadata into the product	Propagated metadata	SAF-P09
Record component lineage (which output versions were used)	Component lineage	CSDA Traceability

NSI Example — Labour Market Publication:

For Q3-2026 the team selects the validated `OD-LFS-2026Q3` employment and unemployment outputs (national and NUTS-2) and the latest administrative vacancy output, pinning their exact versions and carrying their confidence intervals into the integrated product base.

Construct Composite Indicators or Indices

Value created: *Derived measures — indices, ratios, time series — that summarize and add meaning.*

This sub-process computes the designed composite indicators, indices, and time-series aggregates from the integrated components, aligned with GSBPM 7.2. Indicators and indices are calculated per the designed formulae and weighting; time series are chained, linked, and benchmarked per the specification; and each composite carries an appropriate uncertainty measure derived from its components. No component estimate is re-estimated — composites are built from the validated outputs as received.

Activity	Key Outputs	Framework Ref
Calculate composite indicators/indices per designed formulae	Composite indicators	GSBPM 7.2
Construct time series (chain, link, benchmark)	Composite time series	GSBPM 7.2
Derive composite uncertainty from component accuracy	Composite accuracy measures	SAF-P09
Record composition method and inputs as lineage	Composite lineage	SAF-P09

NSI Example — CPI Headline Index:

The headline CPI is composed from the validated item indices using the designed weighting and chaining; the all-items index and its main divisions are constructed, with each carrying its method and the period weights used.

Perform Consistency and Coherence Checks

Value created: *Confidence that the product is internally consistent and coheres with the wider statistical picture.*

This sub-process runs the designed checks, aligned with GSBPM 6.2 (Validate outputs). Internal-consistency checks confirm totals reconcile to components and breakdowns sum to totals; coherence checks compare the product against related statistics and prior releases within the designed tolerances. Discrepancies are dispositioned per policy — accepted, flagged, or returned for review — and explained where significant. No silent adjustment is made: any reconciliation is documented and traceable.

Activity	Key Outputs	Framework Ref
Run internal-consistency checks (totals, breakdowns reconcile)	Consistency results	GSBPM 6.2
Run coherence checks vs. related and prior statistics	Coherence results	GSBPM 6.2
Disposition discrepancies (accept / flag / return) and explain	Discrepancy dispositions	GSBPM 6.2
Log all check results to the product audit trail	Audit trail entries	SAF-P10

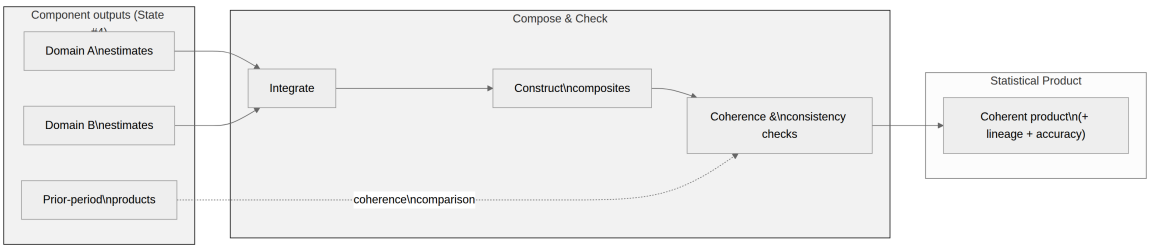


Figure 7: Composition and Coherence Architecture. Component outputs are integrated and composed into indicators, then checked for internal consistency and coherence against prior products before becoming a coherent product carrying its lineage and accuracy.

NSI Example — Business Statistics Product:

The composite SBS product is checked: NACE-class turnover sums to the published total, the survey- and register-based components agree within tolerance for overlapping domains, and the series is coherent with the prior year; one domain exceeds tolerance and is returned to VS-04 for review rather than silently adjusted.

Generate Analytical and Interpretive Outputs

Value created: *Products users can understand — narrative, key messages, and visualizations grounded in the data.*

This sub-process produces the human-facing content, aligned with GSBPM 6.3 (Interpret and explain outputs). The analytical narrative and commentary are written per the templates, key messages are stated with their supporting evidence, and tables and visualizations are produced. Content passes the editorial and quality-review workflow to ensure it is accurate, plain-language, accessible, and consistent with the data and its caveats (including the accuracy carried from VS-04).

Activity	Key Outputs	Framework Ref
Write analytical narrative and commentary per templates	Product narrative	GSBPM 6.3
State key messages with supporting evidence	Key messages	GSBPM 6.3
Produce tables and visualizations	Visualizations & tables	GSBPM 6.3
Run editorial and quality review of content	Reviewed content	GAMSO QM

NSI Example — Labour Market Publication:

The quarterly commentary explains the 0.3pp rise in the employment rate, notes it is within the confidence interval, and presents a regional map and a time-series chart; the editorial review confirms the wording does not overstate a movement that is not statistically significant.

Approve and Register Statistical Products

Value created: *A validated, approved, registered product ready for dissemination.*

The final execution sub-process validates, approves, and registers the product, aligned with GSBPM 6.5 (Finalise outputs) and 7.1. The product is validated against the dissemination and confidentiality standards (including the disclosure risk arising from combining outputs), submitted to the designated approval authority, and — on approval — a persistent immutable product version is created with a unique identifier and full lineage (components, method, coherence results, approval record), registered per SAF-P09, with access permissions applied per SAF-P10. The sub-process concludes by publishing a "Statistical Product Registered" event, notifying downstream VS-06.0 that an approved product is available for dissemination.

Output: Approved Statistical Product (Steady State #4) — the official handover artefact to the dissemination domain.

Activity	Key Outputs	Framework Ref
Validate the product against dissemination and confidentiality standards	Validation result	GSBPM 6.5
Submit to and obtain sign-off from the approval authority	Approval record	SAF 4.1.5

Activity	Key Outputs	Framework Ref
Create an immutable, identified product version with full lineage	Versioned product	CSDA Traceability
Register metadata and publish "Statistical Product Registered" event to notify VS-06.0	Catalog registration, event	SAF-P07

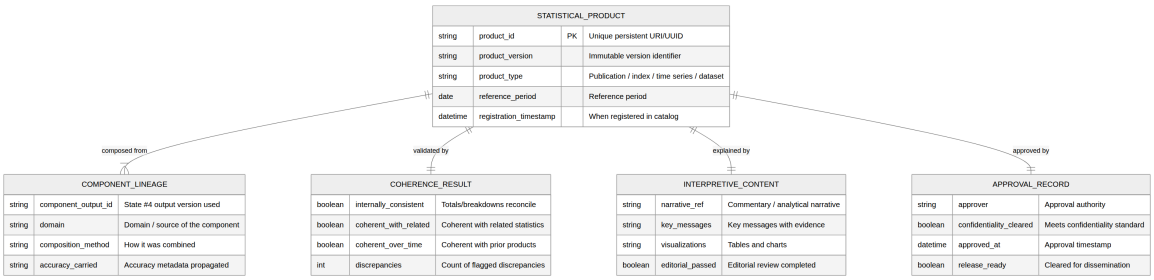


Figure 8: Statistical Product Artefact Structure. Reading guide: each box is a type of record with its fields; a connecting line means the records are linked. Each product version is linked to its component lineage, coherence results, interpretive content, and approval record, all traceable to the State #4 outputs it was composed from.

Management Processes — "provides quality reports"

Framework alignment: GSBPM overarching Quality/Metadata Management — see Framework Mappings.

The management phase runs **alongside and after** execution. It monitors the composition domain's performance, assesses the coherence and interpretability of products, ensures every product is approved and reproducible, and feeds improvement recommendations back into the design phase — closing the governance cycle.

Monitor Process Performance

Operational performance of the composition domain must be tracked systematically across production cycles. The process owner and quality manager collect and aggregate key performance indicators, monitor throughput against the release calendar, and track paradata — which component versions and methods were used, when, by whom, and how many approval iterations occurred. This information feeds process performance dashboards that give the process owner a real-time view of domain health, aligned with the GAMSO Quality Management capability.

Key Performance Indicators (KPIs):

KPI	Description	Measurement
On-time product rate	Products composed within the release-calendar window	Per product, per cycle
Coherence-check pass rate	Products passing coherence checks on first run	Per product, per cycle
Approval-iteration count	Number of approval cycles before sign-off	Per product, per cycle
Component reuse rate	Share of products reusing existing validated outputs	Per cycle
Reproducibility rate	Products regenerable from components + method	Per cycle
Processing time	Duration from output intake to product registration	Per product, per cycle

Activity	Key Outputs	Framework Ref
Collect and aggregate KPIs per production cycle	Process Performance Dashboard	GAMSO QM
Monitor throughput against the release calendar	Throughput Report	GAMSO QM
Track paradata: component versions, methods, approval iterations	Paradata Records	GAMSO QM
Generate process performance dashboards	Dashboard for Process Owner	GAMSO QM

Assess Product Quality

Beyond process performance, the quality of the products themselves must be evaluated against the criteria established in design. Each product is assessed across multiple quality dimensions: coherence (internal consistency and consistency with related and prior statistics), interpretability (clarity and accuracy of narrative and visualizations), completeness (all designed content present), and accuracy propagation (component accuracy correctly carried). Results are compared against the designed quality criteria from Section 3.5. Quality indicators are produced per product and per cycle, and trends are tracked to detect recurring coherence issues or interpretive weaknesses. This aligns with the CSDA Data Quality capability and SAF Capability 2.1.3.

Activity	Key Outputs	Framework Ref
Evaluate coherence (internal and with related/prior statistics)	Coherence Report (per cycle)	SAF 2.1.3
Assess interpretability of narrative and visualizations	Interpretability Review	CSDA Data Quality
Check completeness and correct accuracy propagation	Product Completeness Report	SAF 2.1.3
Identify quality trends across products and cycles	Quality Trend Analysis	SAF 2.1.3

Manage Product and Version Lifecycle

The product portfolio and the versions it produces must remain current, complete, and well-governed. This sub-process governs the lifecycle of product specifications, composite-indicator definitions, and content templates — versioning them, managing transitions (e.g., an index rebasing), and retiring obsolete products. It verifies metadata completeness — that every product carries full component lineage, composition method, coherence results, and approval as an SAF-P09 compliance check — and governs the product-version store: retention, immutability, and reconciliation that catalog entries match stored versions. Metadata lineage from outputs through composition to products is tracked.

Activity	Key Outputs	Framework Ref
Govern product, indicator, and template lifecycle and versions	Product Version Register	SAF-P09
Verify metadata completeness (component lineage, method, coherence, approval)	Metadata Quality Report	SAF-P09
Govern the product-version store: retention, immutability, reconciliation	Version Store Audit	CSDA Traceability
Track metadata lineage from outputs to products	Metadata Lineage Audit	CSDA Traceability

Audit and Compliance Reporting

This sub-process produces the evidence that the composition domain operates within methodological, legal, and organizational requirements. Reproducibility-audit reports demonstrate that every product can be regenerated from its component versions plus the composition method — supporting transparency and accountability. Approval-and-confidentiality reporting confirms that every registered product was approved by the designated authority and meets the confidentiality standard, including the disclosure risk of combining outputs per SAF-P11. Standards compliance reporting confirms conformance to product and exchange standards (e.g., SDMX), and security compliance reporting verifies access controls and the integrity of the approval record. These reports satisfy the SAF Capability 4.1.5 (Accountability) obligation and feed into VS-07.0 Governance and Compliance.

Activity	Key Outputs	Framework Ref
Produce reproducibility-audit reports: every product regenerable	Reproducibility Audit Summary	SAF 4.1.5
Report on approval and confidentiality compliance of products	Approval & Confidentiality Report	SAF-P11
Report on standards compliance (products, SDMX exchange)	Standards Compliance Report	SAF 4.1.5
Report on security compliance: access controls, approval-record integrity	Compliance Report (per cycle)	SAF-P10

Continuous Improvement

The final management sub-process closes the governance loop by analyzing performance and quality data to identify and drive improvements. Quality trends across production cycles are examined: are coherence issues falling, are approval iterations shrinking, is component reuse rising, is interpretability improving? Improvement opportunities are identified — refining composite-indicator methods, improving coherence checks that generate false positives, streamlining the approval workflow, enhancing visualizations, and increasing reuse of validated outputs. These improvement recommendations are fed back to the design processes (Section 3), prioritized based on impact and feasibility, and tracked through subsequent design-implement-execute cycles. This is the SAF Capability 4.1.4 (Improve management) in action, and it connects to the GAMSO concept of transferring mature capabilities from innovation into production.

Activity	Key Outputs	Framework Ref
Analyze quality trends across production cycles	Improvement Recommendations Report	SAF 4.1.4
Identify improvement opportunities: indicator methods, coherence checks, approval, visualization, reuse	Updated Design Inputs	SAF 4.1.4
Feed recommendations back to design processes (Section 3)	Design feedback (closing the governance loop)	SAF 4.1.4
Prioritize improvements based on impact and feasibility	Improvement Tracking Log	GAMSO Capability Mgmt
Track implementation of accepted improvements	Improvement Tracking Log	GAMSO Capability Mgmt

Business Services and Interfaces

The VS-05.0 business domain exposes the following services:

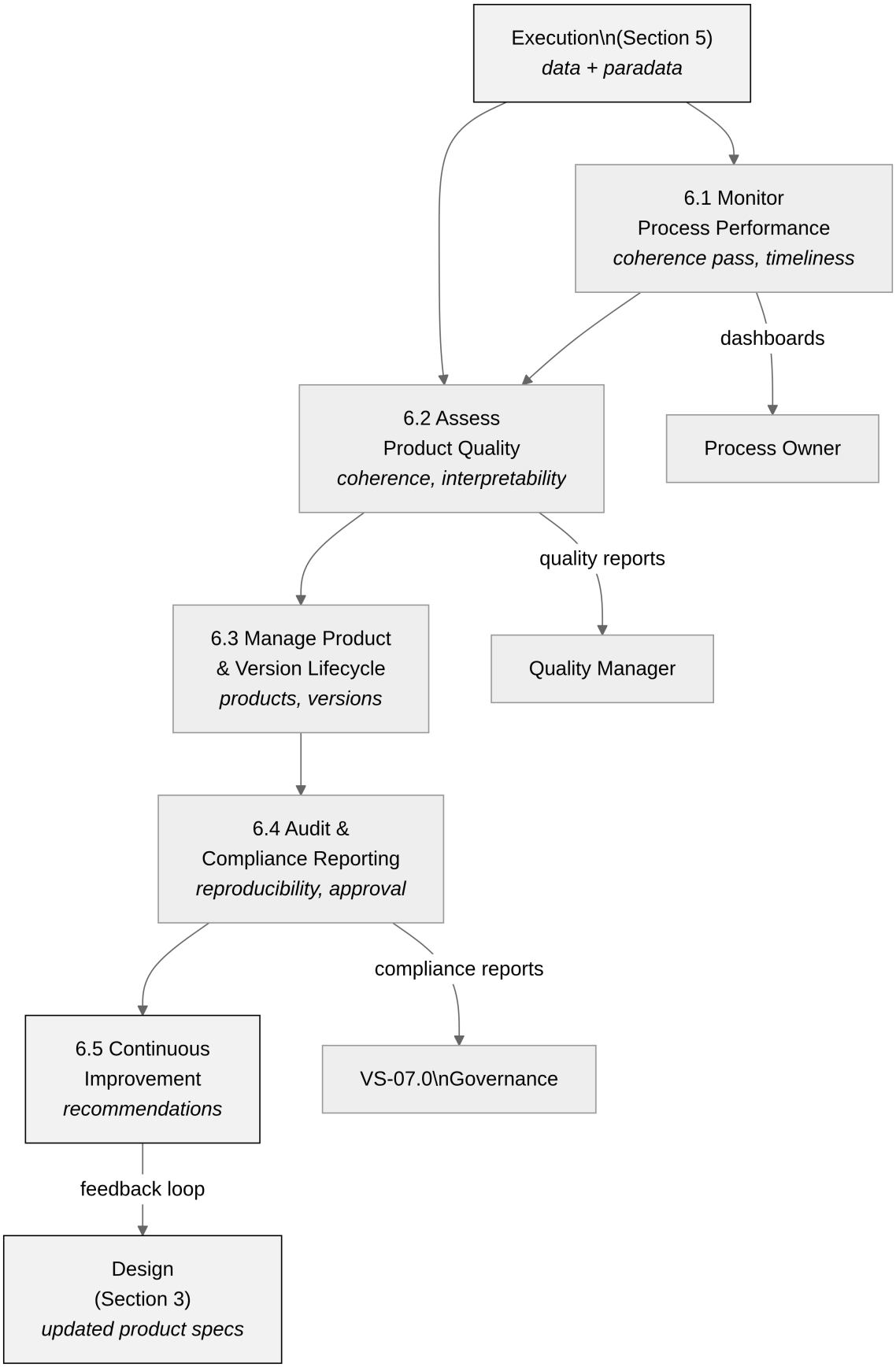


Figure 9: Management Feedback Loop. Execution data flows through monitoring, quality assessment, product/version management, and audit processes. Continuous improvement feeds recommendations back to Design, closing the governance cycle.

Service	Direction	Consumers	Mapped Processes
Integration and Composition Service	Internal (VS-04.0 → NSI)	Output data consumers, composition teams	5.1 Select and Integrate Source Datasets
Indicator and Index Service	Internal	Composition teams, methodologists	5.2 Construct Composite Indicators or Indices
Coherence Validation Service	Internal	Quality managers, subject-matter authors	5.3 Perform Consistency and Coherence Checks
Analytical Content Service	Internal	Authors, editors, downstream dissemination	5.4 Generate Analytical and Interpretive Outputs
Product Provisioning Service	Internal (NSI → VS-06.0)	Dissemination teams	5.5 Approve and Register Statistical Products
Product Catalog Service	Internal	All data consumers across the NSI	6.3 Manage Product & Version Lifecycle
Design and Methodology Service	Internal	Product managers, indicator methodologists	3.1–3.5 All design processes

Interface specifications:

- **Product Provisioning Service** interface: event-driven notification ("Statistical Product Registered") + catalog query API for downstream consumers to discover and access approved products, their lineage, and accuracy
- **Indicator and Index Service** interface: composite-construction API and a methodology/parameters query API
- **Product Catalog Service** interface: search/browse API for products, component lineage, coherence results, and approval records

Boundary Interface Specifications

The domain is loosely coupled (SAF-P07) to its neighbors through two steady-state interfaces. Specifying each boundary explicitly — the handover artefact, the metadata it must carry, the event that triggers it, and where responsibility transfers — makes the loose-coupling contract concrete and testable.

Boundary	Handover Artefact	Required Metadata	Trigger Event	Responsibility Transfer
VS-04.0 → VS-05.0 (intake)	Output Data / Statistics (State #4)	Methodology and parameters, weights, accuracy measures, model diagnostics, applied-SDC record, output-version identifier, lineage to processed source, BIV classification	"Output Data Registered" event	VS-04.0 retains custody of the output version; VS-05.0 assumes responsibility for composing products on event receipt

Boundary	Handover Artefact	Required Metadata	Trigger Event	Responsibility Transfer
VS-05.0 → VS-06.0 (handover)	Approved Statistical Product (State #4)	Component lineage, composition/indicator method, propagated accuracy, coherence results, interpretive content, approval record, confidentiality status, product-version identifier, BIV classification	"Statistical Product Registered" event	VS-05.0 retains custody of the product version; VS-06.0 assumes responsibility for dissemination and service delivery on event receipt

Each interface exchanges a registered, immutable steady-state artefact plus its metadata — never internal working data — so neither side needs knowledge of the other's internal processing.

Business Objects and Data Flow

Business Events

Framework Mappings

This reference table maps each process group of this document to the international frameworks (GSBPM, GAMSO, CSDA) and the SAF capability model. It is intended for architects and auditors; the narrative sections above do not depend on it.

Process group	GSBPM	GAMSO	CSDA	SAF Capabilities
Design	Phase 1 (Specify Needs), Phase 2 (Design: 2.1–2.5)	Production (design activities), Capability Management	Data Governance, Metadata Management, Data Quality	1.1.2 Design statistics product, 2.1.2 Metadata management, 2.1.3 Statistical quality management
Implementation	Phase 3 (Build: 3.1–3.7)	Capability Management	Data Processing / Dissemination preparation (setup), Data Governance (configuration)	1.1.3 Build statistics product, 3.2.1 Application development, 3.2.2 Functional management

Process group	GSBPM	GAMSO	CSDA	SAF Capabilities
Execution	Phase 6 (Analyze: 6.2 Validate outputs, 6.3 Interpret & explain, 6.5 Finalise outputs) + Phase 7 (7.1 Update output systems, 7.2 Produce dissemination products)	Production (execution)	Data Analysis / Dissemination preparation (execution)	2.1.2 Metadata management, 2.1.3 Statistical quality management
Management	Overarching processes (Quality Management, Metadata Management)	Corporate Support (Quality Management, Data Management, Metadata Management), Capability Management (improvement)	Data Quality, Data Governance, Traceability	2.1.3 Statistical quality management, 4.1.4 Improve management, 4.1.5 Accountability

Applying This to Your NSI

Maturity Assessment Checklist

Use this checklist to assess your NSI's maturity across all four process types — not just execution.

Design Maturity

- ☐ Do we have documented product specifications with their component outputs and selection rules?
- ☐ Do we have formal composite-indicator and time-series construction specifications?
- ☐ Do we have documented internal-consistency and coherence checks with tolerances?
- ☐ Do we have content templates and an editorial-review policy for interpretive content?
- ☐ Do we have an approval, confidentiality, and product-versioning policy defined before composition begins?

Implementation Maturity

- ☐ Are our composition engines built from design specifications (not ad-hoc spreadsheets)?
- ☐ Do composites carry an uncertainty measure derived from their components?
- ☐ Can a product be reproduced identically from its component versions plus the composition method?
- ☐ Is the approval workflow and versioning store implemented and tamper-evident?
- ☐ Do we have a formal testing process (technical, coherence, reproducibility)?

Execution Maturity

- ☐ Do we select pinned component output versions and carry their accuracy metadata forward?
- ☐ Do we construct composites without re-estimating component outputs?
- ☐ Do we run internal-consistency and coherence checks and explain discrepancies?
- ☐ Do we produce reviewed interpretive content consistent with the data and its caveats?
- ☐ Do we approve and register every product as an immutable version with full lineage?

Management Maturity

- ☐ Do we track KPIs (coherence pass rate, approval iterations, reproducibility) across cycles?
- ☐ Do we assess coherence and interpretability of products?
- ☐ Do we govern product/version lifecycles and the product-version store?
- ☐ Do we produce reproducibility-audit, approval/confidentiality, and standards-compliance reports?

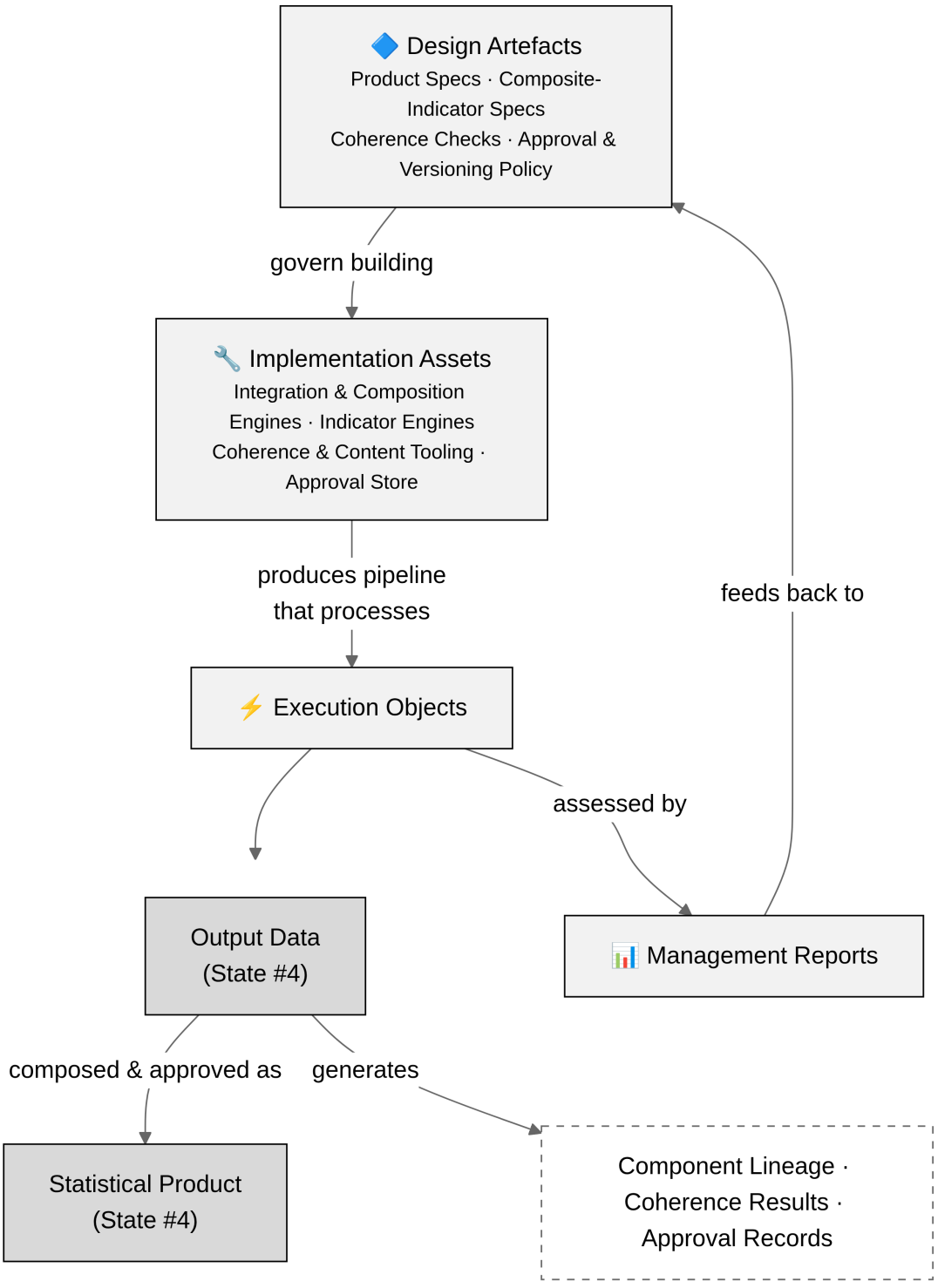
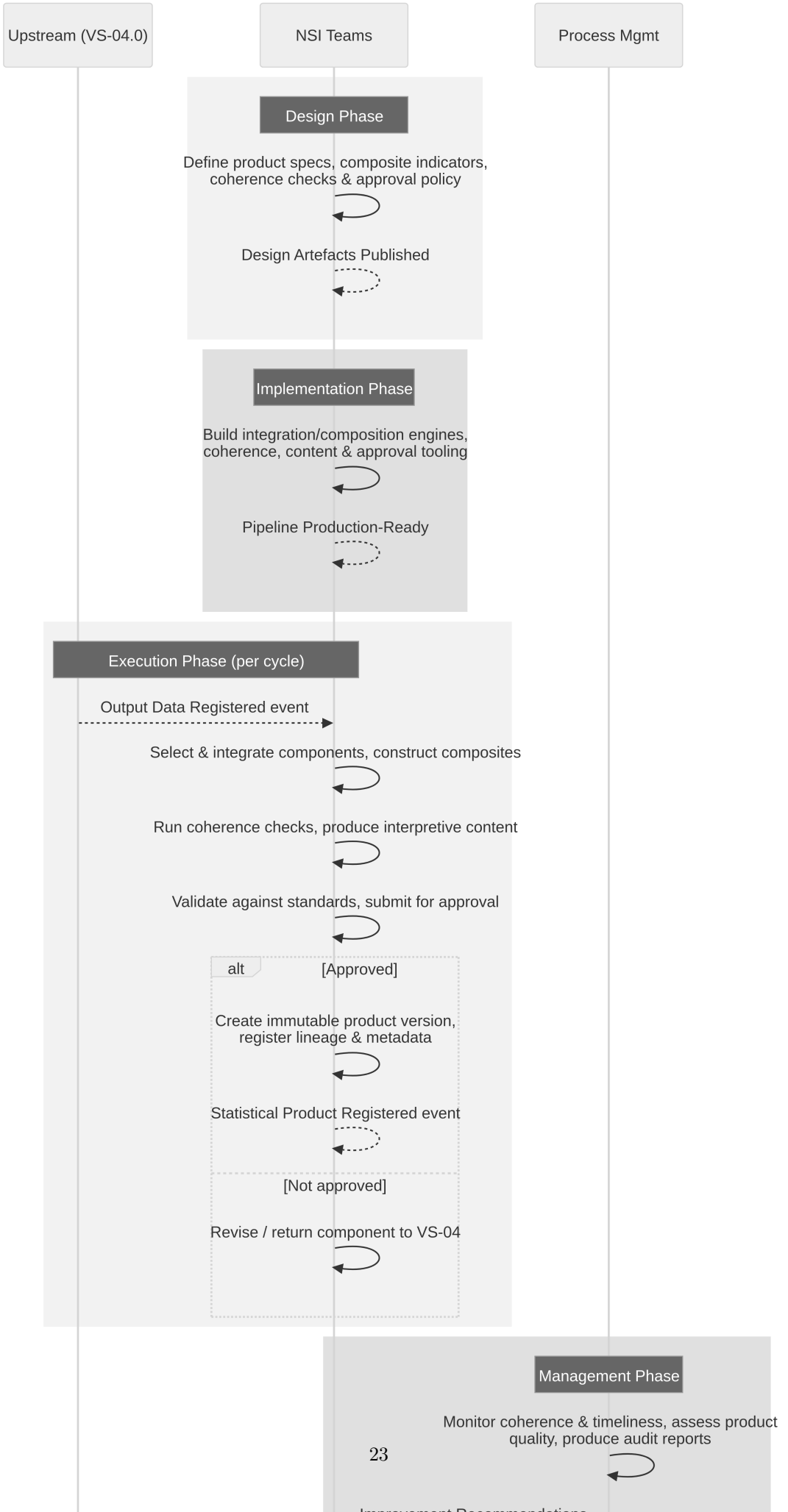


Figure 10: Business Object Lifecycle across all four process types. Design objects govern implementation, which produces the pipeline that composes execution objects into products, which are assessed by management objects, which feed back to design.


Upstream (VS-04.0)
NSI Teams
Process Mgmt
Design Phase
Define product specs, composite indicators, coherence checks & approval policy
Design Artefacts Published
Implementation Phase
Build integration/composition engines, coherence, content & approval tooling
Pipeline Production-Ready
Execution Phase (per cycle)
Output Data Registered event
Select & integrate components, construct composites
Run coherence checks, produce interpretive content
Validate against standards, submit for approval
alt
[Approved]
Create immutable product version, register lineage & metadata
Statistical Product Registered event
[Not approved]
Revise / return component to VS-04
Management Phase
Monitor coherence & timeliness, assess product quality, produce audit reports

☐ Do we have a formal improvement process that feeds back into design?

End-to-End Scenarios

Each scenario walks through **all four process types**, demonstrating how the governance pattern works in practice.

Scenario A: Labour Market Quarterly Publication

Design Phase: Product managers specify the quarterly labour-market publication: employment and unemployment rates (national and NUTS-2), a vacancy indicator, narrative, and a regional visualization, drawing on the validated LFS and administrative outputs. Coherence checks against prior quarters and the approval workflow are designed.

Implementation Phase: Engineers configure the integration engine (pinned LFS/administrative outputs), the time-series engine for the headline series, coherence checks vs. prior quarters, and the content/approval tooling. Coherence and reproducibility tests pass on historical data.

Execution Phase: For Q3-2026, an Output Data Registered event arrives. The team integrates the pinned `OD-LFS-2026Q3` outputs (carrying their CIs), constructs the seasonally-relevant series, and runs coherence checks (all reconcile; coherent with Q2). The commentary explains the 0.3pp employment-rate rise and notes its significance; visualizations are produced and editorially reviewed. The product is approved and registered as an immutable version with full lineage; VS-06.0 is notified.

Management Phase: Coherence-check pass rate is 100%, approval completed in one iteration, all component CIs correctly propagated. Improvement recommendation: add an automated check comparing the vacancy indicator to a related administrative series. Recommendations fed back to design.

Scenario B: Structural Business Statistics Composite Product

Design Phase: The composite SBS product is specified to integrate the register-based and survey-based turnover/employment estimates for the same NACE × region domains, with internal-consistency checks (domains sum to totals) and coherence vs. national-accounts inputs.

Implementation Phase: Engineers configure integration of the pinned `OD-BUS` outputs, the consistency checks, and dominance-aware confidentiality validation for the composite. Coherence and reproducibility tests pass.

Execution Phase: Q3-2026: an Output Data Registered event arrives. The register- and survey-based components are integrated for overlapping domains; NACE-class turnover sums to the published total; the components agree within tolerance except one domain, which is returned to VS-04 for review rather than adjusted. Once resolved, the product is approved and registered (`SP-BUS-2026Q3-001`) with lineage to its components; VS-06.0 is notified.

Management Phase: One coherence discrepancy this cycle (resolved upstream); component reuse rate high; reproducibility audit passes. Improvement recommendation: tighten the tolerance for the recurring domain and document the reconciliation. Recommendations fed back to design.

Scenario C: Consumer Price Index Product

Design Phase: The CPI product is specified to compose the headline all-items index and main divisions from the validated item indices, with chaining/rebasing rules, coherence checks against the prior month and year, and analytical commentary.

Implementation Phase: Engineers configure the index-composition engine (designed weights and chaining), coherence checks, and the commentary/approval tooling. Coherence and reproducibility tests pass; the composed headline matches the expected value on test data.

Execution Phase: For the monthly run, an Output Data Registered event arrives with the validated item indices. The headline and division indices are composed with the period weights, coherence checks confirm consistency with the prior month and year, and the commentary explains the main contributors to the monthly change. The product is approved and registered with full lineage; VS-06.0 is notified.

Management Phase: Coherence pass rate 100%, headline reproducible from item indices + weights, approval in one iteration. Improvement recommendation: enhance the contributions visualization for the commentary. Recommendations fed back to design.